

CUSTOMER CHURN PREDICTION AND ANALYSIS

Project Overview

Customer churn is an important business challenge because losing existing customers can reduce revenue and increase the cost of acquiring replacement customers. This project analyzed customer data to understand patterns associated with churn and developed a machine-learning model to predict customer churn.

The analysis focused on customer service plans, usage behavior, call activity, charges, customer service interactions, and geographic patterns. A Random Forest Classification model was then developed to combine these variables and predict whether customers were likely to churn.

Business Problem

The business needs to better understand customer churn and identify characteristics associated with customers leaving the service.

The project therefore aimed to answer the following questions:

- How common is customer churn?
 - Which service plans are associated with different churn rates?
 - How does customer usage differ between churned and non-churned customers?
 - Are customer service interactions associated with churn?
 - Does churn vary across states?
 - Can customer information be used to predict churn?
 - Which variables contribute most to churn predictions?
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Dataset and Data Quality

The project used the **Churn BigML 80 dataset**, containing **2,666 customer records and 20 variables**.

The dataset includes information about customer location, account characteristics, service plans, call activity, usage, charges, customer service interactions, and churn status.

The initial data-quality assessment found:

- **2,666 records**
- **20 variables**
- **No missing values**
- **No duplicate rows**
- **388 churned customers**
- **2,278 non-churned customers**
- **14.6% overall churn rate**

The target variable was imbalanced because non-churned customers represented the majority of records. Therefore, multiple model evaluation metrics were used instead of relying only on accuracy.

Executive Summary

This project analyzed customer churn using exploratory data analysis and machine learning. The objective was to identify customer characteristics associated with churn and develop a predictive model that could support customer-retention strategies.

The dataset contained **2,666 customers and 20 variables**, with no missing values or duplicate records. **388 customers (14.6%)** had churned, while **2,278 (85.4%)** had not.

Machine Learning Approach

The categorical variables were transformed into numerical representations before modelling.

- International plan → binary encoding
- Voice mail plan → binary encoding
- State → one-hot encoding
- Churn → binary target

The resulting dataset contained **68 features**.

The data was divided into:

- **80% training data:** 2,132 records
- **20% testing data:** 534 records

A **Random Forest Classifier** with 100 trees and `random_state=42` was trained on the training data.

Model Performance

The model achieved an accuracy of **94.38%** and a ROC-AUC score of **0.89**.

For the churn class:

- Precision: **1.00**
- Recall: **0.62**
- F1-score: **0.76**

The confusion matrix showed that the model correctly identified **48 of the 78 customers who actually churned**, while **30 churners were missed**.

This distinction is important because accuracy reflects overall classification performance, while churn recall shows how many actual churners the model successfully identified.

Feature Importance

The three most influential features in the Random Forest model were:

1. **Total day minutes**
2. **Total day charge**
3. **Customer service calls**

International-plan status also contributed meaningfully to the model's predictions.

The feature-importance results support the exploratory analysis, particularly the observed relationship between churn and daytime usage and customer service interactions.

Machine Learning Model

A **Random Forest Classifier** was developed using an 80/20 train-test split. Categorical variables were encoded before modelling, resulting in **68 features**.

The model achieved:

- **94.38% accuracy**
- **0.89 ROC-AUC**
- **1.00 precision for churn**
- **0.62 recall for churn**
- **0.76 F1-score for churn**

The model correctly identified **48 of the 78 actual churners**, while 30 churners were missed.

The three most influential features were:

1. Total day minutes
 2. Total day charge
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Key Findings

Customers with an international plan recorded a substantially higher observed churn rate of **43.70%**, compared with **11.27%** among customers without one.

Churned customers also recorded higher average daytime usage, with **205.18 minutes** compared with **175.10 minutes** among non-churned customers. In addition, churned customers averaged **2.21 customer service calls**, compared with **1.45** among non-churned customers.

Churn rates also varied geographically, with Texas recording the highest observed churn rate among the states shown at **29.09%**.

International call frequency showed a relatively small difference between churned and non-churned customers, suggesting limited standalone explanatory values.

Business Implications

The findings suggest that customer usage behavior, service interactions, and plan characteristics can provide useful signals for churn analysis.

The business could prioritize further investigation of:

- International-plan customers
- High-usage customers
- Customers with frequent customer service interactions
- Regions with higher observed churn rates

The predictive model can be used to support customer-risk monitoring and prioritize retention outreach. However, the missed churn cases show that there is room to improve churn detection, and model predictions should be considered alongside other customer information.

Recommendations

1. Develop a churn-risk monitoring process.
 2. Investigate the reasons for higher churn among international-plan customers.
 3. Review recurring issues among customers with frequent service calls.
 4. Assess whether high-usage customers have suitable service plans.
 5. Monitor regional churn patterns.
 6. Track retention outcomes after interventions.
 7. Review missed churn cases and test model improvements.
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Conclusion

The project demonstrates how data analysis and machine learning can transform customer data into actionable churn insights. The Random Forest model showed strong overall classification performance, while the exploratory analysis identified specific customer behaviors and service characteristics associated with churn.

The results provide a foundation for targeted customer-retention analysis and demonstrate the practical application of Python, exploratory data analysis, and machine learning in business analytics.